WEIGHTINGS OF LOCALLY QUASI-FINITE MORPHISMS AND RELATIVE ZERO-CYCLES

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Allowed standard prerequisites (editorial).

- Weighting definition and diagram: for a locally quasi-finite morphism a weighting is tested on an etale localization diagram with pi : U to V (More on Morphisms, Definition more-morphisms-definition-weighting, canonical tag 0F3A). cited at proof line 1744. Fixes the diagram in which the weighting w_alpha must be shown to have locally constant integral over pi.
- Base-change compatibility of the weighting of a relative zero-cycle: w_{g^*alpha} = w_alpha composed with g' for a morphism g : S' to S of locally Noetherian schemes (relative-cycles chapter, Lemma lemma-weightings-pre, no canonical tag in the snapshot tag table, defined at relative-cycles source line 1671). cited at proof lines 1746 and 1778. First use: reduces part (1) to local constancy on V. Second use: its proof never used that alpha is a relative cycle, so w_alpha = n w persists after base change in part (2).
- Local constancy along smooth morphisms: the support behaviour of relative cycles along a smooth morphism of relative dimension (relative-cycles chapter, Lemma lemma-relative-cycle-smooth, canonical tag 0H5B). cited at proof line 1758. The computed fibre sum is the coefficient of v in pi_*beta, and the lemma makes that function locally constant on V.
- Universal boundedness of fibres: a morphism locally quasi-finite with quasi-compact source has universally bounded fibres (Morphisms chapter, Lemma morphisms-lemma-locally-quasi-finite-qc-source-universally-bounded, canonical tag 03JA). cited at proof lines 1770-1773. Guarantees a sufficiently divisible integer n making the constructed zero-cycles integral.
- Base change of a weighting is a weighting (More on Morphisms, Lemma more-morphisms-lemma-weighting-base-change, canonical tag 0F3B). cited at proof lines 1780-1782. Lets the reduction to the discrete valuation ring case keep the identity w_alpha = n w after base change.
- Specialization lemma for weightings: a weighting of a locally quasi-finite morphism vanishing at all points over a point y vanishes at all points over every specialization y' of y (More on Morphisms, Lemma more-morphisms-lemma-weighting-specialization, no canonical tag in the snapshot tag table,

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defined at more-morphisms source line 22469). cited at proof lines 1795-1797. Kills the difference weighting w' = w_beta - n w, which vanishes over the generic point of the discrete valuation ring, forcing w_beta = n w and beta = alpha.

Source reference. Original definition 0F3A: sources/stacks-chow/tags.txt, source lines 18860–18860. Cited at proof line 1744: fixes the diagram in which the weighting is tested.

Source reference. Original lemma lemma-weightings-pre: sources/stacks-relative-cycles/pinned-original-source.tex, source lines 1671–1671. Cited at proof lines 1746 and 1778: base-change compatibility of the weighting, and the fact its proof never used the relative-cycle property.

Source reference. Original lemma 0H5B: sources/stacks-chow/tags.txt, source lines 21381–21381. Cited at proof line 1758: makes the coefficient function of the pushed-forward cycle locally constant on V.

Source reference. Original lemma 03JA: sources/stacks-chow/tags.txt, source lines 4632–4632. Cited at proof lines 1770-1773: provides the sufficiently divisible integer n.

Source reference. Original lemma 0F3B: sources/stacks-chow/tags.txt, source lines 18861–18861. Cited at proof lines 1780-1782: keeps the identity w_alpha = n w under base change in the reduction.

Source reference. Original lemma more-morphisms-lemma-weighting-specialization: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22469–22469. Cited at proof lines 1795-1797: kills the difference weighting vanishing over the generic point of the discrete valuation ring.

Lemma 0.2. *Let S be a locally Noetherian scheme. Let $f : X \to S$ be a locally quasi-finite morphism of schemes.*

1. *For $\alpha \in z(X/S, 0)$ the map $w_\alpha : X \to \mathbf{Z}$ constructed above is a weighting.*
2. *If X is quasi-compact, then given a weighting $w : X \to \mathbf{Z}$ there exists an integer $n > 0$ such that $nw = w_\alpha$ for some $\alpha \in z(X/S, 0)$.*
3. *The integer n in (2) may be chosen to be a power of the prime p if S is a scheme over $\mathbf{F}_p$.*

Proof. Let $\alpha \in z(X/S, 0)$ and choose a diagram

$$\begin{array}{ccc} X & \xleftarrow{h} & U \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle \pi} \\ Y & \xleftarrow{g} & V \end{array}$$

as in More on Morphisms, Definition 0F3A. Denote $\beta \in z(U/V, 0)$ the restriction of the base change $g^*\alpha$. By the compatibility with base change (Lemma lemma-weightings-pre) we have $w_\beta = w_\alpha \circ h$ and it suffices to show that $\int_\pi w_\beta$ is locally constant on V. Next, note that

$$\begin{aligned} \left(\int_\pi w_\beta\right)(v) &= \sum\nolimits_{u \in U, \pi(u) = v} \beta(u)[\kappa(u) : \kappa(v)]_i[\kappa(u) : \kappa(v)]_s \\ &= \sum\nolimits_{u \in U, \pi(u) = v} \beta(u)[\kappa(u) : \kappa(v)] \end{aligned}$$

This last expression is the coefficient of v in $\pi_*\beta \in z(V/V, 0)$. By Lemma 0H5B this function is locally constant on V.

Conversely, let $w : X \to S$ be a weighting and X quasi-compact. Choose a sufficiently divisible integer n. Let α be the family of 0-cycles on fibres of X/S such that for $s \in S$ we have

$$\alpha_s = \sum\nolimits_{f(x)=s} \frac{nw(x)}{[\kappa(x) : \kappa(s)]_i}[x]$$

as a zero cycle on X_s. This makes sense since the fibres of f are universally bounded (Morphisms, Lemma 03JA) hence we can find n such that the right hand side is an integer for all $s \in S$. The final statement of the lemma also follows, provided we show α is a relative 0-cycle. To do this we have to show that α is compatible with specializations along discrete valuation rings. Note that neither the construction of w_α nor the proof of Lemma lemma-weightings-pre uses that α is a relative cycle. Hence we have $w_\alpha = nw$ and this remains true after base change. As well, the base change of a weighting is a weighting, see More on Morphisms, Lemma 0F3B. Thus we reduce to the problem studied in the next paragraph.

Assume S is the spectrum of a discrete valuation ring with generic point η and closed point 0. Let $w : X \to S$ be a weighting with X quasi-finite over S. Let α be the family of 0-cycles on fibres of X/S constructed in the previous paragraph (for a suitable n). We have to show that $sp_{X/S}(\alpha_\eta) = \alpha_0$. Let $\beta \in z(X/S, 0)$ be the relative 0-cycle on X/S with $\beta_\eta = \alpha_\eta$ and $\beta_0 = sp_{X/S}(\alpha_\eta)$. Then $w' = w_\beta - nw : X \to \mathbf{Z}$ is a weighting (using the result above) and zero in the points of X which map to η. It follows from More on Morphisms, Lemma more-morphisms-lemma-weighting-specialization that $w' = 0$. This means $w_\beta = nw$, hence $\alpha = \beta$ as desired. □